

# Combining Strain and Ferroelectricity in WSe<sub>2</sub> Transistors for Fast, Low-Power Nonvolatile Logic and Multilevel Memory: A Multiscale Simulation Study

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**Abstract**—Strain engineering and van der Waals ferroelectrics have each advanced two-dimensional electronics, but their combination has not been explored. We present an end-to-end multiscale simulation of a p-type ferroelectric field-effect transistor uniting both: a monolayer WSe<sub>2</sub> channel under process-induced biaxial compression, gated through a CuInP<sub>2</sub>S<sub>6</sub> (CIPS) metal-ferroelectric-metal-insulator-semiconductor stack. A two-valley Boltzmann transport model, calibrated against published full-band calculations and strained transistor measurements, feeds self-consistent electrostatics coupling a multidomain Preisach and Landau-Khalatnikov description of CIPS to the strained channel. Compression of 1% more than doubles the hole mobility by suppressing intervalley scattering, and this benefit transfers intact to the nonvolatile domain: the retained on-current rises 1.8 times while the memory window (1.24 V for a 30 nm stack) changes by only about 3%, so strain and polarization act as independent design knobs. Complementary nonvolatile inverters and latches restore their state after complete power loss, cut worst-case static power by eight orders of magnitude versus the reported resistor-loaded latch, and gain 2.3 times in energy-delay product under strain. Strain equally multiplies the sense margins of partially programmed multilevel states, enabling robust two-bit nonvolatile cells. The complete code base is open source.

**Index Terms**—Ferroelectric field-effect transistor (FeFET), CuInP<sub>2</sub>S<sub>6</sub>, multilevel memory, multiscale simulation, nonvolatile logic, strain engineering, two-dimensional (2D) materials, WSe<sub>2</sub>.

## I. INTRODUCTION

**T**WO-DIMENSIONAL (2D) semiconductors are serious candidates for the channel of future nanosheet transistors [1], [2], [3], [4], and monolayer WSe<sub>2</sub> is the leading p-type option because it combines a light K-valley hole mass with contacts that can be doped to low resistance [5], [6]. Two recent developments make WSe<sub>2</sub> p-type field-effect transistors (FETs) especially interesting. First, strain engineering has moved from a curiosity to a quantitative design tool. Process-induced biaxial compression, applied with evaporated stressor films, was shown to raise the hole mobility of monolayer WSe<sub>2</sub> FETs at a rate of  $340 \pm 95\%$  per percent strain, with the on-current rising even faster [7]. A companion full-band transport study showed that this gain comes from the strain-tuned energy separation between the  $\Gamma$  and K valence valleys, which throttles intervalley phonon scattering, and that the mechanism

is robust against temperature, carrier density, impurity density, and the dielectric environment [8]. Both studies end with the same open question: how do these strain gains compose with other device concepts, such as doping, dielectric engineering, and new gate stacks [7], [8]?

Second, van der Waals (vdW) ferroelectrics have matured into practical gate materials. CuInP<sub>2</sub>S<sub>6</sub> (CIPS) retains a switchable out-of-plane polarization at room temperature down to nanometer thickness [9], and CIPS-gated FETs on MoS<sub>2</sub>, InSe, WSe<sub>2</sub>, and MoTe<sub>2</sub> channels show large memory windows and long retention [10], [11], [12], [13], [14]. Most recently, CIPS-based metal-ferroelectric-metal-insulator-semiconductor (MFMIS) ferroelectric FETs (FeFETs) were assembled into the first nonvolatile ferroelectric latch: a circuit that stores its logic state inside the transistors and restores it after complete power loss [14]. That work also exposed two limitations: the demonstrated latch used 10 M $\Omega$  resistor pull-ups, which burn static power of order  $V_{DD}^2/R \approx 1 \mu\text{W}$  per stage and slow the pull-up transition, and all FeFETs were n-type, so a complementary, low-power nonvolatile logic family is still missing. Meanwhile, sliding and interfacial ferroelectricity in WTe<sub>2</sub>, h-BN, and rhombohedral-stacked dichalcogenides is widening the family of switchable vdW gate materials such logic could later adopt [15], [16], [17], [18], [19].

These two threads meet naturally in one device: a p-type MFMIS FeFET whose WSe<sub>2</sub> channel is compressively strained by the stressor process already proven on p-FETs, and whose gate stack is the CIPS/h-BN combination already proven in nonvolatile latches. Strain multiplies the channel conductance while the ferroelectric supplies the stored state; if the two knobs do not interfere, the read current, restore speed, and energy of nonvolatile logic all inherit the full strain gain. No experiment or simulation has yet tested this composition.

In this article we close that gap with an end-to-end, openly available simulation framework that spans from strained valley physics to latch transients: a two-valley Boltzmann transport model of monolayer WSe<sub>2</sub>, anchored to the open C2DB database [20], [21] and calibrated once against the published full-band enhancement curve [8], a multidomain Preisach ensemble with Landau-Khalatnikov (LK) kinetics fitted to measured CIPS parameters [9], [14], a self-consistent electrostatic solver of the MFMIS stack that captures the charge-limited, partially switched states real 2D FeFETs occupy [23], [24], and compact circuit models extracted from the device

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solver. We find that the memory window is set almost entirely by the ferroelectric stack while the read current, delay, and energy are multiplied by strain: the two mechanisms compose rather than trade off. The composition also opens a regime neither technology reaches alone, partially programmed multilevel states whose sense margins are uniformly multiplied by strain. Replacing the resistor-loaded latch with a strained complementary pair improves static power by about eight orders of magnitude and energy-delay product by 2.3 times at  $-1\%$ .

## II. DEVICE CONCEPT AND MULTISCALE FRAMEWORK

Fig. 1(a) shows the simulated device. A monolayer WSe<sub>2</sub> channel sits on an Al<sub>2</sub>O<sub>3</sub>/SiO<sub>2</sub> substrate with doped contacts, as in the strained p-FET experiment of Zhao *et al.* [7]. The gate stack above it follows the MFMIS design of Lee *et al.* [14]: a 10 nm h-BN interlayer, a floating metal gate, a CIPS ferroelectric (30 nm baseline, 86.5 nm in validation runs), and a metal top gate. Biaxial compression  $\epsilon < 0$  is applied as done experimentally with evaporated AlO<sub>x</sub> stressors, which transfer about  $-0.22\%$  today with larger values expected from optimized anchoring [7], [26]. The MFMIS geometry is chosen because the floating metal equalizes the potential across the ferroelectric, gives uniform switching, and suppresses gate leakage [14].

Fig. 1(b) illustrates the transport mechanism. In unstrained monolayer WSe<sub>2</sub> the  $\Gamma$  valley lies only 157 meV below the K valley maximum, so thermally excited holes scatter between the light K valley and the heavy  $\Gamma$  valley through zone-center phonons [8], [22]. Biaxial compression drives the valleys apart at 341 meV per percent strain [8], freezing out this intervalley channel. Fig. 1(c) summarizes the simulation chain; Section III and the supplementary material (Sections S1 to S7) document all layers and parameters.

## III. SIMULATION FRAMEWORK AND CALIBRATION

### A. Two-Valley Strained Transport

The valence band is modeled as a light K/K' valley pair ( $m_K = 0.36 m_0$ , degeneracy 2) and a heavy  $\Gamma$  valley ( $m_\Gamma = 2.2 m_0$ ), with masses and lattice constant from the open C2DB database [20], [21]. The separation  $\Delta E_{\Gamma K}(\epsilon) = 157 \text{ meV} - 341 (\text{meV}/\%) \epsilon$  and the slow K-edge shift of  $-30 \text{ meV}/\%$  follow the first-principles gauges [8], [7]. Scattering rates use Fermi golden rule expressions for 2D parabolic bands [22]: acoustic deformation potential ( $D_{ac} = 2.3 \text{ eV}$  at K, sound velocity 3300 m/s), zero-order optical phonons (31 meV,  $4.6 \times 10^8 \text{ eV/cm}$ ), intervalley phonons (27 meV) with a logistic broadening of 90 meV representing final-state broadening, and screened charged impurities ( $5 \times 10^{12} \text{ cm}^{-2}$ ). The single calibrated quantity is the intervalley coupling  $D_{iv} = 1.5 \times 10^{11} \text{ eV/m}$ , fixed by matching the published full-band enhancement curve at zero strain [8]; sensitivity to this calibration is shown in supplementary Section S2. Mobility follows from the Kubo-Greenwood integral per valley with Fermi-Dirac statistics.

### B. Ferroelectric Model

CIPS is a Preisach ensemble of square hysterons with Gaussian coercive-field spread ( $\sigma = 22\%$  of  $E_c = 0.3 \text{ MV/cm}$ ), remanent polarization  $5.5 \mu\text{C/cm}^2$ , and background permittivity 25 [9], [14]. Kinetics use the LK equation per domain with Landau coefficients set by  $(P_r, E_c)$  and a viscosity calibrated to the measured 60  $\mu\text{s}$  programming pulse [14], [23].

### C. Self-Consistent MFMIS Solver and Circuit Models

For each gate drive the solver enforces  $x = t_{FE} E_{FE} + D/C_{hBN} + \psi_s(p)$  with  $D = \epsilon_0 \epsilon_b E_{FE} + P_{sw}$  and  $p = D/q$ , where  $\psi_s$  inverts the 2D Fermi-Dirac hole density of the K valley. Hysterons flip in small batches, lowest coercive field first, until the internal field is below every remaining threshold, which yields the self-limited minor-loop states that govern retention [24]. Read current uses the linear-region drift model with 50 k $\Omega \mu\text{m}$  contacts, within the 1 to 370 k $\Omega \mu\text{m}$  span reported for TOS-doped contacts [7]. Circuit projections use smooth all-region compact models with thresholds extracted from the solver, a 3 V supply, 1 fF node capacitance, and explicit transient integration. Full equations, parameters, and convergence details are in supplementary Sections S1 to S5.

## IV. RESULTS AND DISCUSSION

### A. Strain-Enhanced Hole Transport, Validated Against Full-Band Theory and Experiment

Fig. 2 presents the transport layer. Fig. 2(a) shows the strain dependence of the  $\Gamma$ -K separation adopted from the first-principles gauge of Afridi *et al.* [8]. Fig. 2(b) shows the computed total K-valley scattering rate versus hole energy. The model includes acoustic deformation potential, zero-order optical, intervalley, and screened charged-impurity scattering, evaluated with Fermi golden rule expressions for 2D parabolic valleys (Section III). Under compression the intervalley contribution collapses and the total rate at 300 K drops by almost an order of magnitude; under tension the  $\Gamma$  valley overtakes the K valley and scattering grows.

Fig. 2(c) is the central validation of this layer. With a single calibrated intervalley coupling (Section III), the model reproduces the full-band enhancement on the entire compressive branch: we obtain  $\mu_h/\mu_{h0} = 1.44$  at  $-0.25\%$ , 1.86 at  $-0.5\%$ , and 2.30 at  $-1\%$ , against published full-band values of 1.40, 1.75, and 2.37 [8], with a baseline mobility of 24.9  $\text{cm}^2/\text{Vs}$  that matches the reported extrinsic value of about 25  $\text{cm}^2/\text{Vs}$ . The model also falls inside the uncertainty band of the strained-FET measurement at  $-0.22\%$  [7]. On the tensile side the reduced-order model overestimates the degradation beyond  $+0.5\%$  because the heavy  $\Gamma$  valley takes over completely (Fig. 2(d)); our device operates only in compression, and this branch is documented in supplementary Section S2. The key point for what follows is that the mobility gain is a channel property: it does not depend on how the gate charge is supplied, so it should survive under a ferroelectric gate.

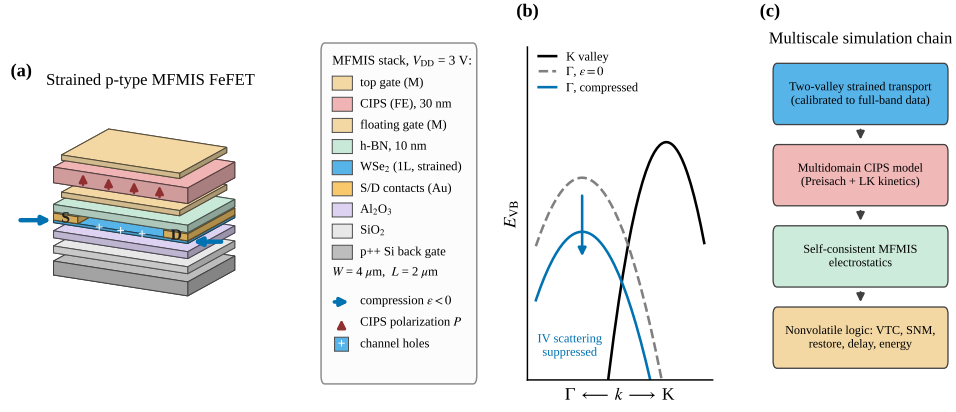


Fig. 1. Device concept and simulation chain. (a) Exploded 3D view of the strained p-type MFMIS FeFET: a compressively strained monolayer WSe<sub>2</sub> channel (holes marked +) with source (S) and drain (D) contacts, h-BN interlayer, floating gate, CIPS ferroelectric with switchable out-of-plane polarization  $P$  (red arrows), and top gate. (b) Valence band schematic. Compression lowers the  $\Gamma$  valley away from the K valley maximum, which suppresses intervalley (IV) phonon scattering of holes. (c) The four coupled layers of the open-source simulation framework.

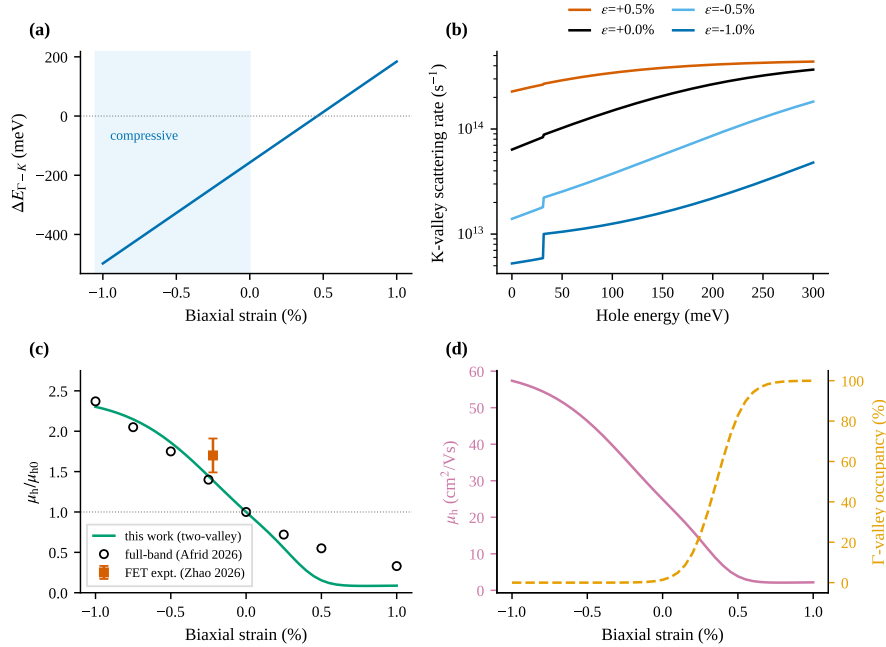


Fig. 2. Strain-dependent hole transport in monolayer WSe<sub>2</sub>. (a)  $\Gamma$ -K valence valley separation versus biaxial strain, following the first-principles gauge of 341 meV/% [8]. (b) Total K-valley scattering rate versus hole energy for four strains. (c) Normalized hole mobility from our two-valley model (line) compared with the published full-band results (circles, [8]) and the strained-FET experiment (square with error bar, [7]). (d) Absolute mobility (left axis) and  $\Gamma$ -valley hole occupancy (right axis) versus strain at  $p = 10^{13} \text{ cm}^{-2}$ ,  $n_{\text{imp}} = 5 \times 10^{12} \text{ cm}^{-2}$ .

### B. Ferroelectric Gate Stack

Fig. 3 presents the CIPS layer of the framework, built as a Preisach ensemble of 400 to 1600 domains with a Gaussian spread of coercive fields around  $E_c = 0.3 \text{ MV/cm}$ , each carrying the measured remanent polarization  $P_r = 5.5 \mu\text{C/cm}^2$  and a linear background permittivity of 25 [9], [14]. The quasi-static loop (Fig. 3(a)) reproduces the measured  $P_r$  by construction and gives realistic rounded minor loops (supplementary Section S3). Domain kinetics follow the LK equation with a viscosity calibrated once against the measured 60  $\mu\text{s}$  programming pulse of the CIPS MFMIS FeFET [14]. Figs. 3(b) and (c) show the resulting switching transients and

the switching time versus overdrive: full reversal takes 71  $\mu\text{s}$  at  $1.5 E_c$  and accelerates steeply with field, consistent with the measurable current modulation observed for programming pulses as short as 1  $\mu\text{s}$  [14]. Because polarization switching is orders of magnitude slower than the electrostatic response of the channel, the FeFET sweeps below treat the ensemble quasi-statically, and the LK time scale enters only the write operation of the logic circuits.

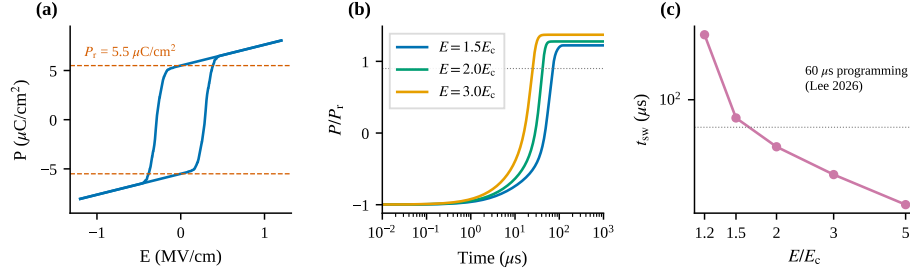


Fig. 3. Multidomain CIPS model. (a) Quasi-static P-E loop of the Preisach ensemble with  $P_r = 5.5 \mu\text{C}/\text{cm}^2$  and  $E_c = 0.3 \text{ MV}/\text{cm}$ . (b) LK switching transients from negative saturation for three field overdrives. (c) Time to 90% polarization reversal versus overdrive; the dotted line marks the  $60 \mu\text{s}$  programming pulse reported for the experimental device [14].

### C. The Strained p-Type FeFET: Strain Multiplies the Read Current but Leaves the Memory Window Untouched

The device solver couples the two layers above. For each gate bias it enforces displacement continuity through the stack with zero net charge on the floating gate, computes the hole density from the  $\text{WSe}_2$  2D density of states, and lets hysteresis flip one small batch at a time until the internal field drops below every remaining coercive field (Section III). This self-limited update matters: because  $P_r$  of CIPS exceeds the maximum channel charge by more than an order of magnitude, the stable retained states are partially switched minor-loop states whose polarization matches the charge the channel can actually supply, exactly the regime real 2D FeFETs occupy [14], [23], [24]. A model that forced full polarization reversal would predict depolarization fields far above  $E_c$  and immediate state loss; ours yields a retained depolarization field of  $0.56 E_c$  for the 30 nm stack (supplementary Section S4), safely inside the stability margin.

Fig. 4(a) shows double-sweep transfer curves of the baseline device (30 nm CIPS, 10 nm h-BN,  $\pm 6 \text{ V}$  drive). The loops are counterclockwise in the p-FET convention, as observed for CIPS-gated channels [10], [14], with a constant-current memory window of 1.24 V, an on-current of tens of  $\mu\text{A}$  for a  $4 \mu\text{m}$  wide channel, and a nonvolatile on/off ratio, read at zero gate bias after program and erase, of  $5.7 \times 10^7$ . This matches the several-orders-of-magnitude contrast reported experimentally [14]. Scaling the CIPS thickness (Fig. 4(b)) moves the window from 0.6 V at 15 nm to 4.4 V at the 86.5 nm experimental thickness, in agreement with the 2 to 4 V windows seen in measured CIPS FeFETs [10], [12], [14], at the cost of a proportionally larger program voltage (supplementary Section S4). The 30 nm design point keeps the full logic operable at  $V_{\text{DD}} = 3 \text{ V}$ .

Figs. 4(c) and (d) contain the first main result. Compressive strain raises the retained on-current from  $5.7 \mu\text{A}$  to  $10.5 \mu\text{A}$  at  $-1\%$ , a gain of 1.8 times, while the memory window changes by only about 3% across the whole strain range. The gain is slightly below the bare mobility ratio of 2.3 because the contact resistance takes a growing share of the read voltage as the channel improves, an effect the solver includes explicitly. The window is strain-immune because it is set by the coercive voltage of the CIPS layer and the charge balance of the stack, neither of which involves the channel

mobility; the small residual variation comes from the strain-induced K-valley shift, which moves both threshold voltages together by about 30 meV per percent strain [7]. Strain and polarization therefore act as orthogonal design knobs: one sets the speed and energy of the read, the other sets the stored-state separation. This is the device-level confirmation of the multiplicative composition conjectured in the strain studies [7], [8].

### D. Complementary Nonvolatile Logic: Restore After Power Loss at a Fraction of the Static Power

The experimental CIPS latch used n-type FeFETs with  $10 \text{ M}\Omega$  resistor pull-ups because no p-type partner existed [14]. Our device fills that role. We extract compact all-region transistor models from the device solver (threshold voltages  $-0.80 \text{ V}$  and  $+0.43 \text{ V}$  in the gate convention for the erased and programmed states, gain factor proportional to  $\mu_h(\epsilon)$ ) and build a complementary nonvolatile inverter: the strained p-FeFET as pull-up, a normally-off  $\text{WSe}_2$  n-FET as pull-down, mirroring the n-FETs already used in the experimental latch [14].

Fig. 5(a) shows the voltage transfer characteristics. The complementary stage swings rail to rail in both stored states, and the two states displace the switching threshold by about 0.5 V, which is what makes the output state recoverable. Cross-coupling two such inverters gives the butterfly curves of Fig. 5(b) with a 666 mV static noise margin at  $V_{\text{DD}} = 3 \text{ V}$ , comfortable for robust latching. The restore experiment of the CIPS latch is reproduced in Fig. 5(c): both supply rails are removed, all node voltages collapse to zero, and on power-up the latch returns deterministically to its stored state within about 1 ns, driven only by the threshold asymmetry retained in the two FeFETs. This is the same intrinsic restore behavior demonstrated experimentally [14], now without a single resistor.

The complementary topology changes the power picture qualitatively: the resistor-loaded stage dissipates  $V_{\text{DD}}^2/R = 0.9 \mu\text{W}$  whenever its output is low, while the complementary stage leaks only through a subthreshold transistor, worst case  $4.6 \times 10^{-3} \text{ pW}$  in our model, about eight orders of magnitude lower even if interface leakage raises the real number by several decades. Strain then acts on speed and energy: Fig. 5(d) shows the pull-up propagation delay falling from 50 ps to 22

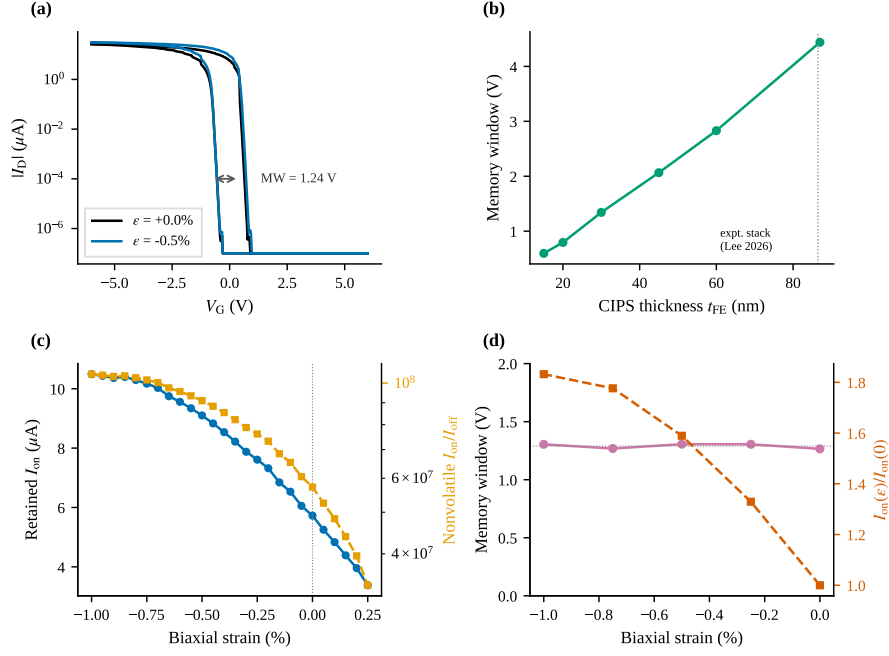


Fig. 4. Simulated strained p-type MFIS FeFET. (a) Double-sweep transfer characteristics ( $\pm 6$  V drive,  $V_{\text{DS}} = -1$  V) at zero and  $-0.5\%$  strain; the counterclockwise hysteresis gives a memory window of 1.24 V. (b) Memory window versus CIPS thickness; the dotted line marks the 86.5 nm experimental stack of [14]. (c) Retained on-current at zero gate bias (left) and nonvolatile on/off ratio (right) versus strain. (d) Memory window (left) and retained on-current gain (right) versus strain: the window stays flat to within about 3% while the read current rises 1.8 times.

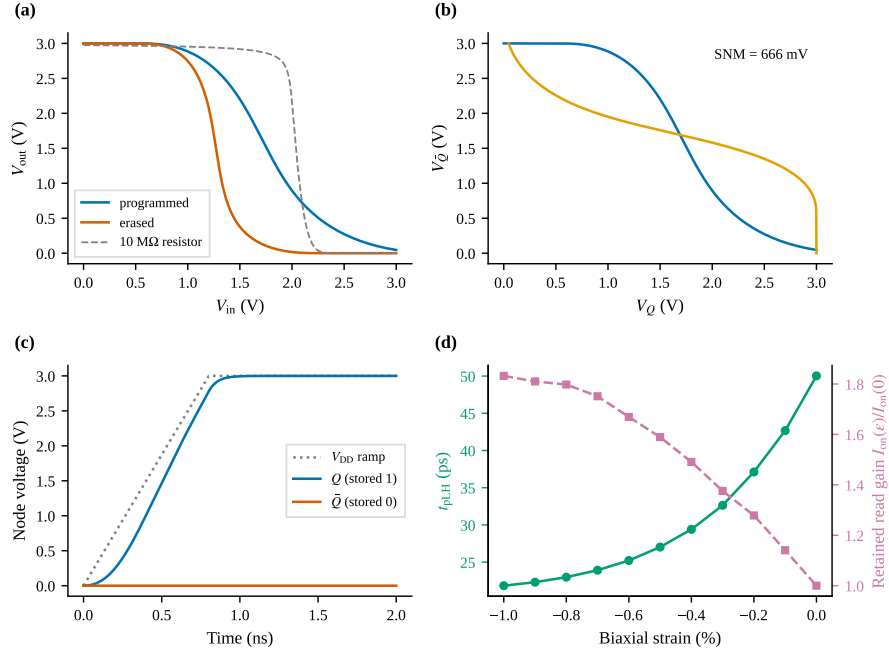


Fig. 5. Strain-augmented complementary nonvolatile logic at  $V_{\text{DD}} = 3$  V. (a) Inverter transfer curves for the two stored FeFET states, compared with a 10 M $\Omega$  resistor-loaded stage of the type demonstrated experimentally. (b) Butterfly plot of the cross-coupled pair; the static noise margin is 666 mV. (c) Power-up restore transient: with complementary programmed states stored in the two pull-ups, the latch nodes  $Q$  and  $\bar{Q}$  return to their pre-power-loss values with no external restore circuitry. (d) Propagation delay (left) and retained read-current gain (right) versus strain:  $-1\%$  compression cuts the delay from 50 ps to 22 ps.

ps as compression reaches  $-1\%$  (1 fF load), with the energy-delay product improving by the same 2.3 times since the switching energy  $CV_{\text{DD}}^2$  is strain-independent. The retained read current gain of 1.8 times applies directly to the restore

and read paths of the latch. Because polarization switching, not the channel, limits the write, the write time stays at the tens-of-microseconds scale of CIPS kinetics (Fig. 3(c)) and is unaffected by strain; applications that write rarely but

TABLE I  
SUMMARY OF THE STRAIN-FERROELECTRIC CO-DESIGN AT  $V_{DD} = 3$  V. EXPERIMENTAL BASELINE VALUES FROM [14] AND [7]

| Quantity                        | This work (30 nm CIPS, $\epsilon = -1\%$ )    | Experimental baseline                   |
|---------------------------------|-----------------------------------------------|-----------------------------------------|
| Memory window                   | 1.24 V (3 V logic compatible)                 | 2 to 4 V (86.5 nm CIPS)                 |
| Nonvolatile on/off              | $5.7 \times 10^7$                             | about $10^5$ to $10^7$                  |
| Retained read current           | $10.5 \mu\text{A}$ ( $1.8\times$ strain gain) | $\mu\text{A}$ scale, no strain          |
| Static power per stage          | $\sim 5 \times 10^{-3}$ pW (complementary)    | $0.9 \mu\text{W}$ (resistor load)       |
| Inverter delay (1 fF)           | 22 ps                                         | resistor-limited, $\sim 10 \mu\text{s}$ |
| Energy-delay product            | $2.3\times$ better at $-1\%$                  | reference                               |
| Multilevel (2-bit) sense margin | $3.6 \mu\text{A}$ ( $1.8\times$ strain gain)  | 4 levels shown, no strain               |
| State restore after power loss  | yes, $< 2$ ns, no extra circuitry             | yes (resistor latch)                    |

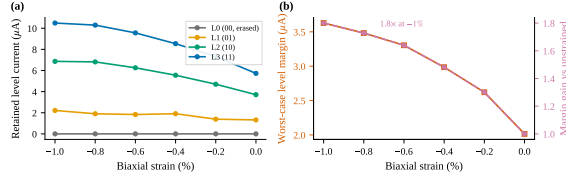


Fig. 6. Strain-multiplied multilevel operation. (a) Four retained conductance levels (two bits per cell) versus biaxial strain, programmed by partial polarization switching. (b) Worst-case adjacent-level sense margin (left) and its gain over the unstrained cell (right): compression multiplies every level separation by the same 1.8 times factor.

read and restore often, such as power-gated processors and intermittently powered edge nodes, benefit most [14], [25].

#### E. A New Operating Regime: Strain-Multiplied Multilevel States

Partial programming turns the FeFET into a multilevel cell, as demonstrated experimentally for CIPS devices [14]. Fig. 6(a) shows four retained states (two bits per cell) written with program drives of  $-6$ ,  $3.5$ ,  $4.3$ , and  $6$  V. Because every level current is the product of a strain-immune retained polarization state and the strain-multiplied channel mobility, compression scales the entire staircase uniformly: the worst-case adjacent-level margin grows from  $2.0 \mu\text{A}$  to  $3.6 \mu\text{A}$  at  $-1\%$  strain, the same 1.8 times factor found for the binary read current (Fig. 6(b)). This is a regime neither parent technology reaches alone. Strain engineering by itself has no stored states to separate, and unstrained multilevel FeFETs must trade sense margin against level count [14], [25]. Strain instead acts as a global gain knob on the whole staircase, converting a marginal two-bit cell into a robust one at unchanged programming conditions, a directly testable prediction of the co-design.

Table I collects the headline numbers of the co-designed device against the experimental baseline. All values come from the same self-consistent framework, and every parameter is traceable to a published measurement or an open database [7], [8], [14], [20].

#### F. Robustness of the Co-Design: Traps, Scaling, and Demonstrated Strain Levels

Three factors could in principle bound these conclusions, and we address each quantitatively (supplementary Section S6). First, the largest compression demonstrated to date is  $-0.22\%$  [7], though TMD monolayers elastically sustain several percent [7], [26]. The co-design does not wait for

$-1\%$ : at  $-0.22\%$  the retained current, energy-delay product, and multilevel margins already improve by 1.29, 1.39, and 1.28 times, and at  $-0.5\%$  by 1.59, 1.85, and 1.57 times. Second,  $\text{WSe}_2/\text{h-BN}$  interface traps are treated two ways: a bounding quasi-static model (instantaneous capture, no re-emission while the channel conducts) and energy-resolved Shockley-Read-Hall kinetics with detailed-balance emission, whose only added parameter is the capture cross section. Both treatments agree that the memory window is essentially trap-immune: it changes by less than about 6% up to  $D_{it} = 10^{12} \text{ cm}^{-2}\text{eV}^{-1}$ , invariant across sweep durations of  $10^{-4}$  to  $10^2$  s and cross sections of  $10^{-16}$  to  $10^{-14} \text{ cm}^2$ , because trapped charge and the ferroelectric minor loop compensate self-consistently. The retained current sets the limit instead: the kinetic model settles at  $3.4 \mu\text{A}$  at  $10^{12}$  and still  $1.7 \mu\text{A}$  (on/off  $1.7 \times 10^7$ ) at  $3 \times 10^{12} \text{ cm}^{-2}\text{eV}^{-1}$ , where the quasi-static bound already predicts collapse, with the strain gain intact at 1.85. Reported h-BN interfaces ( $5 \times 10^9$  to  $10^{11} \text{ cm}^{-2}\text{eV}^{-1}$  [27]) leave more than an order of magnitude of margin; only  $\text{SiO}_2$ -class interfaces near  $10^{13} \text{ cm}^{-2}\text{eV}^{-1}$  [27] destroy the device in both treatments. Third, short-channel electrostatics: with the monolayer  $\text{WSe}_2$  and h-BN permittivities [28], the electrostatic scale length of the stack is  $\lambda = \sqrt{(\epsilon_{\text{ch}}/\epsilon_{\text{hBN}}) t_{\text{ch}} t_{\text{hBN}}} \approx 5.1 \text{ nm}$ , so the long-channel compact models hold for gate lengths down to roughly  $10\lambda \approx 50 \text{ nm}$ , far below the  $2 \mu\text{m}$  operating point; in still shorter channels the strain mechanism itself survives, with uniaxial compression retaining about half the biaxial gain [7], [8]. None of these factors touches the central scaling result, which rests only on the orthogonality of the mobility and polarization knobs. The framework is directly reusable for emerging sliding ferroelectrics, whose ultrathin switchable dipoles [16], [18], [19] could replace CIPS in the same stack at lower voltage.

#### V. CONCLUSION

We have presented, to our knowledge, the first study that composes process-induced compressive strain with a van der Waals ferroelectric gate in a single self-consistent device and carries the composition through to circuit metrics. Compression multiplies the retained read current of a CIPS/ $\text{WSe}_2$  p-type FeFET by 1.8 times at  $-1\%$  while leaving the 1.24 V memory window unchanged to within about 3%: strain and polarization are orthogonal design knobs. A complementary nonvolatile inverter built from this device restores its state after complete power loss in under 2 ns, reduces worst-case static power by about eight orders of magnitude relative to the demonstrated resistor-loaded CIPS latch, and gains 2.3

times in energy-delay, and strain multiplies the worst-case sense margin of four-level, two-bit cells by the same 1.8 times, a strain-multiplied multilevel regime unavailable to either parent technology. These conclusions are robust down to the wire: interface-trap simulations spanning a worst-case bound and finite-rate Shockley-Read-Hall kinetics keep the window essentially unchanged and retention logic-compatible up to  $D_{it} = 10^{12} \text{ cm}^{-2} \text{ eV}^{-1}$  in both treatments, and to  $3 \times 10^{12}$  in the kinetic one, more than an order of magnitude above reported h-BN interfaces; the compact models hold to about 50 nm gate length; and every gain scales continuously from the  $-0.22\%$  compression already demonstrated, where the read current, energy-delay product, and multilevel margins already improve by 1.3 to 1.4 times. All models are calibrated against published measurements of the constituent technologies, and the full code base is openly available, providing a quantitative target for the strained-FeFET experiments that both parent technologies are now ready to attempt.

#### CODE AVAILABILITY

The complete Python code base (all model layers plus the driver that regenerates every figure of this article) is provided as supplementary material and is openly available on [GitHub](#); code, benchmark data, and figure files are permanently archived at DOI: [10.5281/zenodo.22084359](https://doi.org/10.5281/zenodo.22084359).

#### SUPPLEMENTARY MATERIAL

The supplementary PDF contains all derivations, calibration and sensitivity analysis, and additional ferroelectric, retention, trap, and circuit results (Sections S1 to S7, Figures S1 to S7, Tables S1 to S4), plus a code archive with reproduction instructions.

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